



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 02

Aim: To understand and demonstrate the use of basic commands in different operating systems (Windows, Linux, and UNIX) for managing files, directories, permissions, and user interactions through a terminal or command-line interface.

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❓ **Aim:** To understand and demonstrate the use of basic commands in different operating systems (Windows, Linux, and UNIX) for managing files, directories, permissions, and user interactions through a terminal or command-line interface.

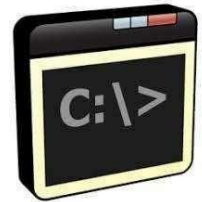
❓ **Objectives:**

1. To learn and practice fundamental command-line operations for file and directory management.
2. To explore and utilize user and permission management commands effectively.
3. To enhance system administration skills by working with commands across different operating systems.

❓ **Requirements:**

Hardware Requirements:

- Processor: Multi-core CPU, Intel Core i3 (3.0 GHz) or higher
- RAM: Minimum 4 GB (8 GB recommended for optimal performance)
- Storage: 100 GB HDD or SSD (Solid State Drive) for faster access
- Network Interface: Ethernet or Wi-Fi adapter for connectivity



Software Requirements:

- Operating System: Windows 10/11, Linux (Ubuntu 20.04/CentOS 8), UNIX-based OS
- Command-line Interface: PowerShell or Command Prompt (Windows), Terminal (Linux/UNIX)
- Text Editor: Nano, Vim, or Visual Studio Code for file editing
- Administrative Privileges: Superuser (Linux/UNIX) or Administrator (Windows) access

❓ **Theory:**

In system administration, command-line interfaces (CLI) are essential tools for managing and interacting with operating systems like Windows, Linux, and UNIX. Commands allow users to perform various tasks such as navigating directories, managing files, controlling permissions, and monitoring system performance. Each operating system provides a set of builtin commands, such as 'man', 'ls', 'cd', 'mkdir', and 'chmod', to facilitate efficient system management. Understanding these commands and their syntax is crucial for automating tasks, enhancing security, and ensuring optimal system functionality. This practical aims to develop foundational skills in executing and applying basic commands across different platforms.

?? Commands:

1. Display User Manual of a Command

- Functionality: Shows the manual page with details about a command's usage, options, and arguments.
- Syntax: `man <command>`
- Example: `man ls`

2. Change Current Working Directory.

- Functionality: Changes the terminal's current working directory.
- Syntax: `cd <directory-path>` - Example: `cd /home/user/Documents.`

3. List Contents of the Current Directory.

- Functionality: Lists all files and directories in the current location.
- Syntax: `ls`
- Example: `ls`

4. Read/Modify/Concatenate Text Files.

- Functionality: Displays or manipulates file content.
- Syntax:
- Read: `cat <filename>`
- Modify: `'nano <filename>`
- Concatenate: `cat <file1> <file2> > <outputfile>`

5. Create a New Directory.

- Functionality: Creates a new directory at the specified path. -
- Syntax: `mkdir <directory-name>` - Example: `mkdir newdir`

6. Display Current Working Directory.

- Functionality: Prints the current directory path.
- Syntax: `pwd`
- Example: `pwd`

7. Write Arguments to Standard Output.

- Functionality: Prints the provided string or variables.
- Syntax: echo <arguments>
- Example: echo Hello World

OUTPUT

```

Command Prompt
Microsoft Windows [Version 10.0.26200.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\amrut>cd OneDrive

C:\Users\amrut\OneDrive>cd Desktop

C:\Users\amrut\OneDrive\Desktop>pwd
/c/Users/amrut/OneDrive/Desktop

C:\Users\amrut\OneDrive\Desktop>date
The current date is: 22-01-2026
Enter the new date: (dd-mm-yy) 22-01-2026
A required privilege is not held by the client.

C:\Users\amrut\OneDrive\Desktop>time
The current time is: 19:40:34.55
Enter the new time:

C:\Users\amrut\OneDrive\Desktop>mkdir demo

C:\Users\amrut\OneDrive\Desktop>touch file.txt

C:\Users\amrut\OneDrive\Desktop>ls
Backend          'MinGW Installer.lnk'    assets                learning.html
CM24006_practical1.xlsx  Miscallounce             demo                 'library system.mwb'
'CodeTantra SEA.lnk'    MongoDBCompass.lnk      desktop.ini          mingw-get-setup.exe
'DBMS Notes.pdf'       RestfullAPI             'er dig.mwb'         mondodb
Dev-C++.lnk           'Visual Studio Code.lnk' file.txt              'my code'
Documents            'Zoom Workplace.lnk'    index2.css           'my expriences.txt'
'MAJORPROJECT(ap)'     'ap(project)'           intenshipform.html   pk.c

C:\Users\amrut\OneDrive\Desktop>pwd
/c/Users/amrut/OneDrive/Desktop

C:\Users\amrut\OneDrive\Desktop>|

```

student@student-BY-OEM: ~

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
student@student-BY-OEM:~$ pwd
/home/student
```

```
student@student-BY-OEM:~$ ls
```

Desktop Documents Downloads Music Pictures Public snap Templates Videos

```
student@student-BY-OEM:~$ cd Document
```

bash: cd: Document: No such file or directory

```
student@student-BY-OEM:~$ mkdir test
```

```
student@student-BY-OEM:~$ rmdir test
```

```
student@student-BY-OEM:~$ touch file.txt
```

```
student@student-BY-OEM:~$ cp file.txt backup.txt
```

```
student@student-BY-OEM:~$ mv old.txt new.txt
```

my: command not found

```
student@student-BY-OEM:~$ rm file.txt
```

```
student@student-BY-OEM:~$ cat file.txt
```

cat: file.txt: No such file or directory

```
student@student-BY-OEM:~$ head file.txt
```

head: cannot open 'file.txt' for reading: No such file or directory

```
student@student-BY-OEM:~$ ps
```

PID	TTY	TIME	CMD
3492	pts/0	00:00:00	bash
5878	pts/0	00:00:00	ps

```
student@student-BY-OEM:~$
```


student@student-BY-OEM: ~

student@student-BY-OEM: \$ head file.txt

head: cannot open 'file.txt' for reading: No such file or directory

```
student@student-BY-OEM:~$ ps
  PID TTY          TIME CMD
 3492 pts/0    00:00:00 bash
 5878 pts/0    00:00:00 ps
```

student@student-BY-OEM:~\$ prtsc

prtsc: command not found

student@student-BY-OEM:~\$ PrtSc

PrtSc: command not found

student@student-BY-OEM:~\$ shift + PrtSc

bash: shift: +: numeric argument required

```
student@student-BY-OEM:~$ ps
  PID TTY          TIME CMD
 3492 pts/0    00:00:00 bash
 5919 pts/0    00:00:00 ps
```

student@student-BY-OEM:~\$ kill 1234

bash: kill: (1234) - No such process

student@student-BY-OEM:~\$ man ls

student@student-BY-OEM:~\$

```
student@student-BY-OEM:~$ ps
  PID TTY          TIME CMD
 3492 pts/0    00:00:00 bash
 5947 pts/0    00:00:00 ps
```

student@student-BY-OEM:~\$ chmod 700 file.txt

chmod: cannot access 'file.txt': No such file or directory

```
student@student-BY-OEM:~$ ps
  PID TTY          TIME CMD
 3492 pts/0    00:00:00 bash
 5949 pts/0    00:00:00 ps
```

student@student-BY-OEM:~\$

8. Remove a File.

- Functionality: Deletes a specified file.
- Syntax: `rm <filename>` - Example: `rm file.txt`

9. Delete a Directory.

- Functionality: Removes an empty directory.
- Syntax: `rmdir <directory-name>`
- Example: `rmdir olddir`

10. Copy a File or Directory.

- Functionality: Copies a file or directory to a destination.
- Syntax: `cp <source> <destination>`
- Example: `cp file.txt backup/`

11. Switch to Root User.

- Functionality: Gains root privileges temporarily.
- Syntax: `sudo su`
- Example: `sudo s`

12. Move Files or Directories.

- Functionality: Moves or renames files and directories.
- Syntax: `mv <source> <destination>`
- Example: `mv file.txt newdir/`

13. Search for a String in a File.

- Functionality: Searches for a specific word or pattern in a file.
- Syntax: `grep "<string>" <file>` - Example: `grep "error" log.txt`

14. Print Top N Lines of a File.

- Functionality: Displays the first N lines of a file.
- Syntax: `head -n <N> <file>` - Example: `'head -n 10 file.txt'`

15. Print Last N Lines of a File.

- Functionality: Displays the last N lines of a file.
- Syntax: `tail -n <N> <file>`

16. Remove Read Permission from Owner.

- Functionality: Revokes the owner's read permission for a file.
- Syntax: `chmod u-r <filename>`
- Example: `chmod u-r file.txt`

17. Change Specific Permissions.

- Functionality: Sets or removes specific file permissions.
- Syntax: `chmod u+r,w-x,g+w <filename>`
- Example: `chmod u+r,w-x,g+w file.txt`

18. Add Write Permission to Owner, None to Others.

- Functionality: Allows write access for the owner only.
- Syntax: `chmod u+w,o-rwx <filename>`
- Example: `chmod u+w,o-rwx file.txt`

19. Assign Permissions to Users.

- Functionality: Modifies file access for users, groups, and others.
- Syntax: `chmod u+rwx,g+rx,o+r <filename>`
- Example: `'chmod u+rwx,g+rx,o+r file.txt`

20. Assign R/W/X to Others.

- Functionality: Gives read, write, and execute permissions to others.
- Syntax: `chmod o+rwx <filename>`
- Example: `chmod o+rwx file.txt`

21. Remove All Permissions from All Users.

- Functionality: Clears all permissions on a file.
- Syntax: `'chmod a-rwx <filename>`
- Example: `'chmod a-rwx file.txt`

22. Remove Read Permission Using Absolute Mode.

- Functionality: Uses numeric mode to restrict read access.
- Syntax: `chmod 700 <filename>`
- Example: `chmod 700 file.txt`

23. Set R/W for Owner, None for Group/Other.

- Functionality: Assigns permissions in numeric mode.

- Example: `chmod u+x,g+r,o+r file.txt`

25. Add Execute Permission to All Users.

- Functionality: Enables execution by everyone.
- Syntax: `chmod a+x <filename>`
- Example: `chmod a+x script.sh`

❑ Conclusion: In conclusion, understanding and using essential operating system commands like 'ls', 'cd', 'cp', 'mv', and 'chmod' enables efficient file management, navigation, and permission control. Tools like 'grep', 'head', and 'tail' enhance data processing. Mastery of these commands improves system administration, task automation, and overall system security and performance.

❑ Discussion Questions:

1. What is the significance of the `pwd` command in a Linux environment?
 2. Explain the function of the `cp` command and its common options.
 3. How does `chmod 700` affect file permissions, and what does each digit represent?
 4. Describe the difference between `head` and `tail` commands in Linux.
 5. What is the purpose of the `grep` command, and how is it used with regular expressions?
- Department of Emerging Technologies, SBJITMR, Nagpur.

☞ References:

<https://ubuntu.com/tutorials/command-line-for-beginners#1-overview>
<https://www.geeksforgeeks.org/25-basic-ubuntu-commands/>

Date:_____

Signature
Course Coordinator
B.Tech CSE(AIML)
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